CS 5593 – Team Project:

Checkpoint 1: First Results and Plan Revision

25 points in total · Due: 10/18 11:59 PM CST

Checkpoint 1 is the first test of whether your original proposal survives contact with the data. By this point, your team should have completed the work you committed to in the Checkpoint 1 section of your proposal and produced preliminary results using real data.

Your report should explain what you planned, what you actually implemented, what you found, and what those findings imply for the next stage of the project.

Progress Against Proposal (2 pts, 2-3 sentences) Briefly restate what your proposal said would be completed by Checkpoint 1 (1 pt), and explain what has actually been completed (1 pt). If your implementation differs from the original proposal, briefly explain why. A justified change will not be penalized simply because the plan changed.

Method Implemented (7 pts) Describe the analysis you have actually completed so far. Your description should include:

- Data used (1 pt). Identify the dataset, subset, records, or variables actually used in the checkpoint analysis.
- Preprocessing / representation (2 pts). Explain how the raw data were cleaned, filtered, transformed, aggregated, normalized, represented, or converted into features for analysis.
- Analysis method (2 pts). Explain the analysis, algorithm, model, or comparison that was actually performed.
- Output logic (1 pt). Make clear what each major step takes as input and what it produces as output.
- Connection to the research question (1 pt). Explain why the implemented method helps address the original research question or hypothesis.

You may now name specific algorithms, models, libraries, or tools. However, do not simply list what you used. Explain the logic of the workflow clearly enough that another person can understand.

Preliminary Results (9 pts) Present concrete results produced from real data. Your results section should include:

- Result artifact(s) (3 pts). Include at least one meaningful result, such as a table, figure, visualization, summary statistic, feature distribution, clustering result, group comparison, or preliminary model output.
- Result interpretation (3 pts). Explain what the result shows and what pattern, relationship, or behavior you observe.
- Connection to the research question or hypothesis (2 pts). Explain why the result matters for the question your project is trying to answer.
- Limitations or uncertainty (1 pt). Identify an important limitation, unexpected result, uncertainty, or unresolved issue revealed by the current analysis.

Do not only show that your code runs. The result should provide interpretable evidence about the problem you proposed to study. At this stage, the results do not need to fully answer your final research question.

Evaluation of the Original Hypothesis and Plan (4 pts) Based on the preliminary evidence, evaluate whether the assumptions and first steps of your original proposal still appear reasonable. Your evaluation should:

- Connect the evidence back to the original proposal (2 pts). Explain whether the current results support, challenge, or leave uncertain the assumptions behind your original hypothesis or analysis plan.
- Identify what still appears valid or needs reconsideration (1 pt). State which part of the original proposal you currently believe should be preserved or reconsidered.
- Interpret the evidence appropriately (1 pt). Avoid making conclusions stronger than what your preliminary results support.

You are not expected to fully confirm or reject your final hypothesis at Checkpoint 1. The goal is to determine whether the current direction still makes sense.

Next-Step Plan or Revision (3 pts) Explain what your team will do next. Your plan should include:

- Next step (1 pt). Identify the specific analysis or task that will be completed next.
- Justification (1 pt). Explain why this next step follows from your current results.
- Continue or revise (1 pt). State whether you will continue with the original plan or revise some part of the project.

If the current evidence supports your proposed direction, explain the next analysis step and what result or artifact you expect to produce. If the current evidence does not support the original plan, explain what you will revise and why. Possible revisions may include the hypothesis, dataset or subset, representation, comparison, analysis method, or interpretation of the research question. A revision is not a failure. The purpose of the checkpoint is to use evidence to decide whether to continue, refine, or change the original plan.

Submission: Submit **one** report per **team** as **a PDF.**

Your report should be organized using the five sections above and should include all figures, tables, and preliminary results needed to support your discussion.

The strongest submissions will not simply show that work was completed. They will show that the team can interpret the evidence and use it to decide what should happen next.